CAMERON WILDING

Bedford, NH | cameronpwilding@gmail.com | (603)-325-0045

advun.github.io | linkedin.com/in/cameron-wilding-3x2/| github.com/advun

PROFESSIONAL SUMMARY

Electrical and Computer Engineering graduate student specializing in digital hardware design and hardware security. Experienced leading multi-person tape-out teams, developing and verifying RTL in Verilog/SystemVerilog, and applying Python/CUDA tooling to hardware-focused AI research. Seeking full-time hardware design or security engineering roles beginning January 2027.

EDUCATION

Worcester Polytechnic Institute, Worcester, MA
Master of Science in Electrical and Computer Engineering *May 2025 - Dec 2026*
GPA: 4.0/4.0
Bachelor of Science in Electrical and Computer Engineering *Aug 2022 - May 2026*
GPA: 3.70/4.0

WORK EXPERIENCE

Research Assistant, Worcester Polytechnic Institute *Jan 2026 – Present*

- Conducting research on cryptographic AI governance by developing a system to verify large language model integrity without access to model weights or internals, as part of M.S. thesis
- Implementing zk-SNARK proof pipelines and behavioural probe frameworks in Python to produce compact, verifiable attestations of model behaviour

Teaching Assistant, Worcester Polytechnic Institute *Aug 2025 – Jan 2026*

- Supported 31 students in ECE3829: Advanced Digital Systems Design with FPGAs
- Guided development of digital circuits and state machines on the Xilinx Artix-7 in Vivado
- Graded lab work and provided technical support in Verilog

PROJECTS

Power Management IC Design for IoT Applications, WPI Major Qualifying Project *Aug 2025 – May 2026*

- Led digital design for an 8-person team developing a custom ASIC using the Cadence tool suite
- Designed control state machines in Verilog to manage power monitoring and sequencing
- Won second place among all WPI Major Qualifying Projects; taped out May 2026, with testing planned for Fall 2026

Trace Data Compressor, Independent *April 2026*

- Designed a lossless data compressor in SystemVerilog to compress bus data for storage on BRAM
- Achieved a 7.4x compression rate on FPGA, winning Best Rookie Hack at GoatHacks 2026
- Taped out with TinyTapeout in April 2026

Tensor Co-Processor for Croc SoC, Independent *Oct 2025 – Dec 2025*

- Designed systolic array, control logic, and partial product accumulation in SystemVerilog
- Optimized for low area and power, with INT8 inputs accumulating to INT32 for high accuracy
- Verified with UVM in December 2025; taped out April 2026, with testing planned for Fall 2026

CubeSat Support Systems Development, WPI Satellite Development Club *Jan 2024 – Dec 2025*

- Led the Support Systems Subteam of the WPI Satellite Development Club
- Developed a C codebase to control the power board, camera, and data transmission
- Designed the main board in Altium to connect and control the power board and camera over SPI

TECHNICAL SKILLS

Hardware/Software: Vivado, KiCad, LTspice, Altium, Git, Cadence Tool Suite (Virtuoso, Genus, Innovus, Tempus, Spectre), SPICE, Linux, UVM, PyTorch, FPGA, LaTeX

Programming Languages: SystemVerilog, Verilog, C, C++, Python, MATLAB